

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Replacer idler plate, Part Number 3164762
- Handle, Part Number 3163089
- Guide pins, Part Number 3164764
- Eye bolts, Part Number 3164767
- Reusable plug, Part Number 4918875
- Gasket scraper or cleaning pad, Part Number 3823258
- Rear crankshaft seal replacer, Part Number 3164780
- Mounting tool, Part Number ST-1325
- Indicator, Part Number 3376050

Additional Service Items

- Slide hammer

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

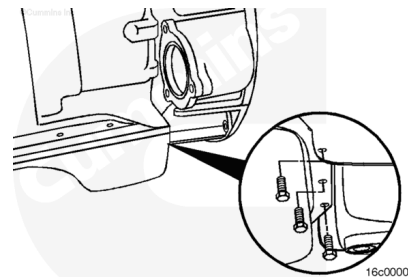
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries or air supply to the starter to prevent accidental engine starting. See the equipment manufacturer service information.
- Drain the oil. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-025.html)
- Remove the transmission, clutch, and all related components. See the equipment manufacturer service information.
- Remove the flywheel. Refer to Procedure 016-005 in Section 16.
(/qs3/pubsys2/xml/en/procedures/377/377-016-005.html)
- Remove the starter motor. Refer to Procedure 013-020 in Section 13.
(/qs3/pubsys2/xml/en/procedures/377/377-013-020-tr.html)

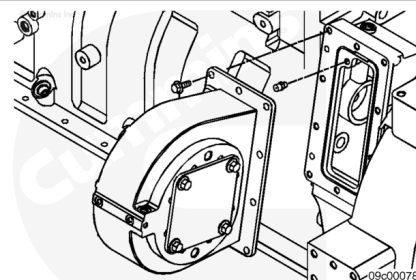
Remove

Remove the capscrews from the rear of the oil pan.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the 10 mm capscrews from the hydraulic housing.

Remove the hydraulic housing from the rear engine power take off (REPTO) flywheel housing.

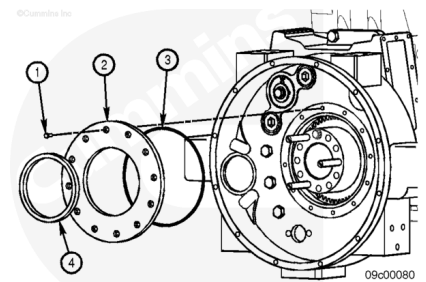
Remove the spray nozzle from the oil supply line hole.

Remove the capscrews (1) from the seal carrier.

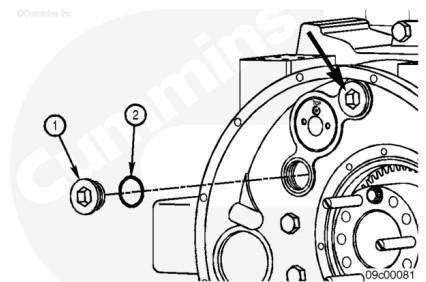
Remove the seal carrier (2) from the REPTO flywheel housing.

Remove oil seal (4) and discard. Refer to Procedure 001-024 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-024.html>)

Remove the o-ring (3) from the groove on the REPTO flywheel housing.



Remove the plugs (1) and o-rings (2) from the REPTO flywheel housing.

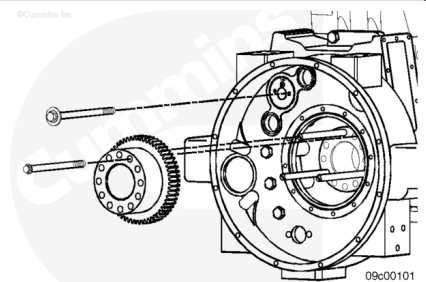


Remove the crankshaft gear.

Remove the idler gear shaft capscrew and washer.

Note : Not all engines will be built with a washer. The idler gear shaft without a washer will feature a 19 mm hole through the shaft.

Remove the guide pins.

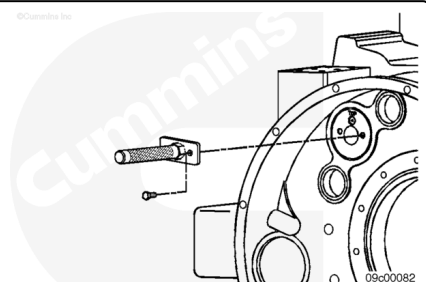


If the idler gear shaft can **not** be removed with the handle, use a slide hammer similar to Snap-On, Part Number CJ114.

Position the replacer idler plate, Part Number 3164762, on the idler gear shaft. Install and tighten the idler gear capscrews. Install the handle, Part Number 3163089.

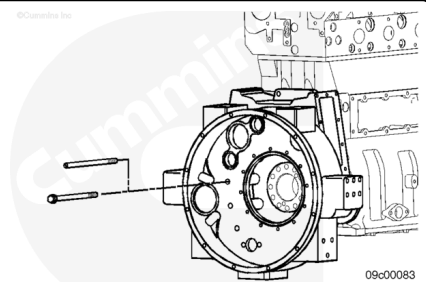
Torque Value: 14 n•m [124 in-lb]

Remove the idler shaft, idler gear, and bearing from the REPTO flywheel housing.

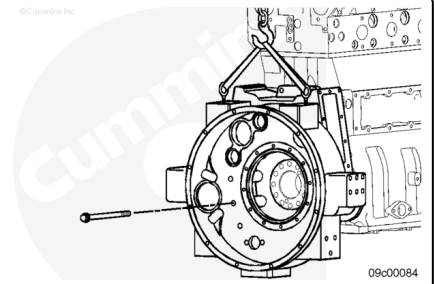


Remove capscrews from the REPTO flywheel housing.

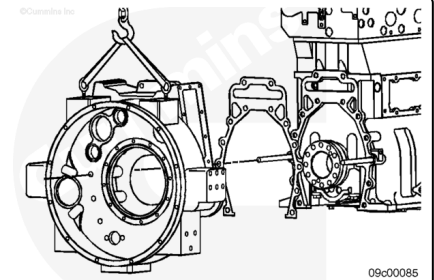
Install the guide pins, Part Number 3164764.



Install the eye bolts, Part Number 3164767, on the top of the REPTO flywheel housing. Install the lifting slings to the eye hooks. Adjust the hoist until there is tension in the lifting sling. Remove the remaining capscrews.

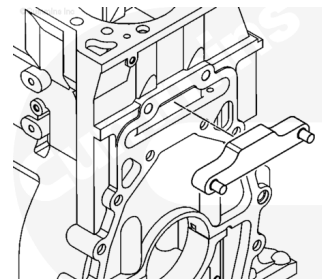


Use a mallet to tap the REPTO flywheel housing off the locating dowels. Remove and discard the gasket.



⚠ CAUTION ⚠

Immediately upon removal of the flywheel housing assembly, a plug must be inserted into the cylinder block oil passage drillings. Failure to insert the oil passage plug can result in a bearing failure, crankshaft failure, or both.



The reusable plug, Part Number 4918875, is necessary to prevent debris from entering the lubricating system during the repair.

Plug the cylinder block oil passage with plug, Part Number 4918875.

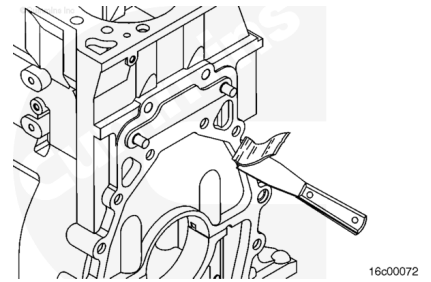
Firmly push the plug into the oil passage to prevent all debris from entering the lubricating system when cleaning the gasket surface.

Clean and Inspect for Reuse

⚠ CAUTION ⚠

Do not allow dirt or foreign material to enter oil passages in the cylinder block when cleaning the gasket sealing surfaces. Main or connecting rod bearing failures can be

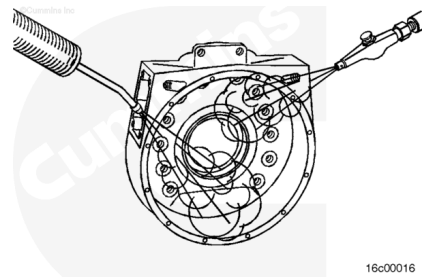
caused if debris is introduced into the lubrication system. Therefore, use of power tools combined with abrasive pads to clean gasket surfaces is not recommended.



Clean the gasket surfaces by hand with a gasket scraper or cleaning pad, Part Number 3823258. If the residue gasket material can **not** be felt with a finger, the surface is ready to accept the new gasket.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

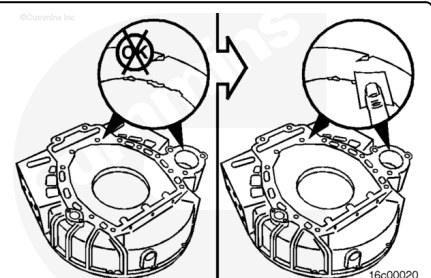
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use steam or solvent to clean the REPTO flywheel housing.
Dry with compressed air.

Inspect all surfaces for nicks, burrs, or cracks.
Use a crocus cloth to remove small nicks and burrs.

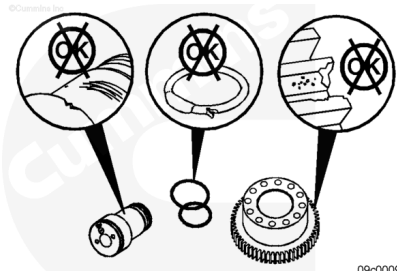


Inspect all threaded capscrew holes for damage.
Repair or replace the housing if the capscrew holes are damaged.



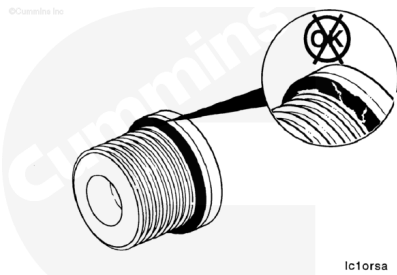
16c00032

Inspect the idler shaft, bearing, and gear for nicks, burrs, or cracks. Replace if damaged.



09c00098

Inspect the o-rings from the plugs and the one large o-ring from the REPTO flywheel housing for cuts or tears. Replace the o-rings if damaged.

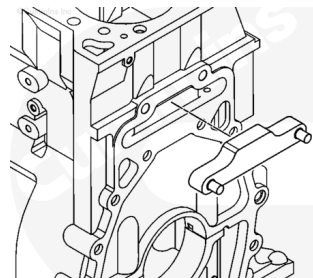


1c10rsa

Install

Remove the protective plug from the block oil passages. Make sure no debris enters the lubrication system.

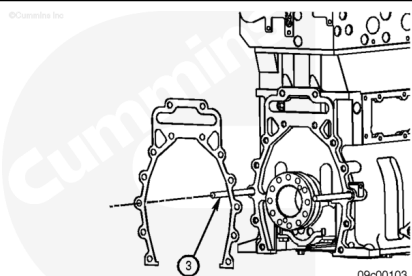
Use a clean shop towel to carefully wipe out the oil passage and inspect to make sure no debris is left in the passage.



16c00071

Install the guide pins (3), Part Number 3164764, into the cylinder block.

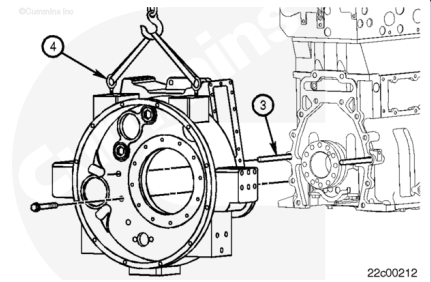
Position the flywheel housing gasket to the rear of the cylinder block using guide pins.



09c00103

⚠ WARNING ⚠

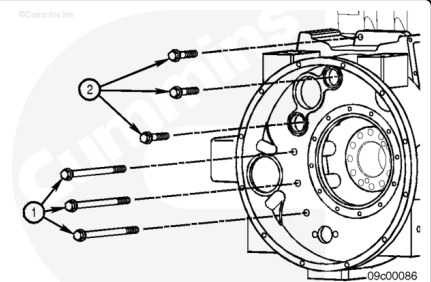
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the eye hooks (4) in the top of the REPTO flywheel housing. Install the lifting slings to the eye hooks and push the REPTO flywheel housing onto the guide pins (3).

Install the short capscrews (2) and long capscrews (1) in the locations shown.

Remove the guide pins and install the other long capscrews.

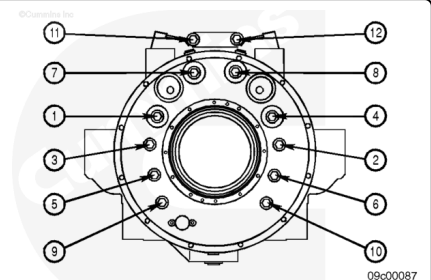


Tighten capscrews (1), (2), (9) and (10).

Torque Value: 7 n•m [62 in-lb]

Tighten the remaining capscrews in the sequence shown.

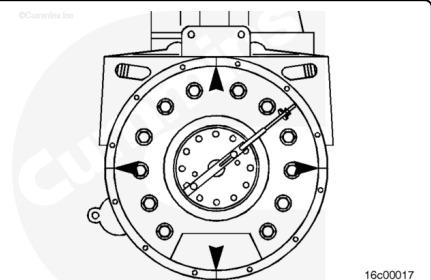
Torque Value: 7 n•m [62 in-lb]



⚠ CAUTION ⚠

The REPTO flywheel housing bore and face must be in alignment with the crankshaft to prevent possible damage to the engine, the clutch, or the transmission.

Use the measure step in this procedure to align the housing.

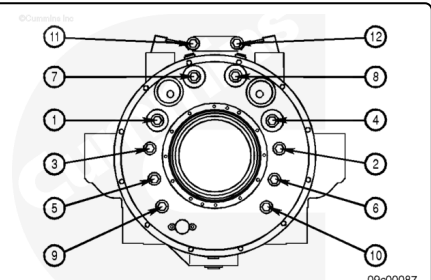


Tighten the capscrews.

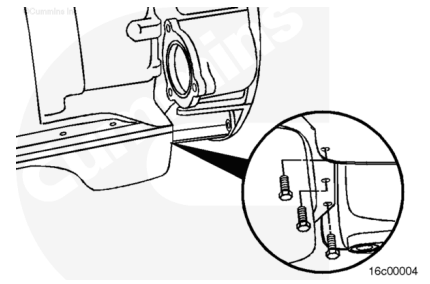
Torque Value:

1. 100 n•m [74 ft-lb]

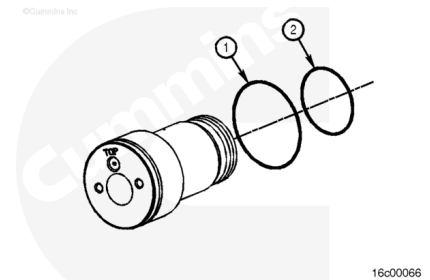
2. 197 n•m [145 ft-lb]



Install the oil pan capscrews.



Install the large o-ring seal (1) and the small o-ring seal (2) onto the idler shaft.

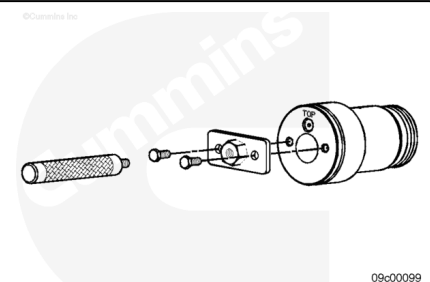


Assemble the replacer idler shaft plate and handle.

Install the capscrews in the idler shaft holes and tighten.

Use clean engine oil to lubricate the idler shaft.

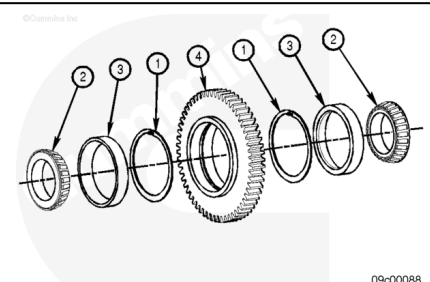
Torque Value: 14 n•m [124 in-lb]



The cones (2) are loose and will fall out if **not** contained.

Make sure the gear (4), bearing snap rings (1), and the cups (3) are pressed in place before inserting the assemblies through the center or sides of the REPTO flywheel housing openings.

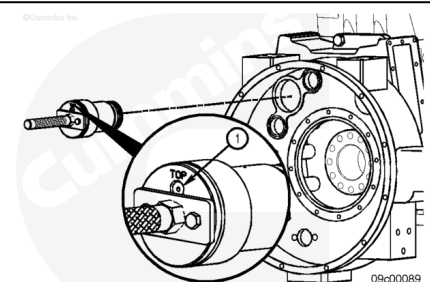
Use clean engine oil to lubricate the bearing snap rings, cones, cups, idler shaft, and gear.



Make sure the idler shaft plug (1) is facing upward when installing it into the bore.

Install the gear and bearing assembly through the center or sides of the REPTO flywheel housing holes. Position the bearing races together and insert the idler shaft with the tool through the first bear bore. The running clearance is 0 to 0.025 mm [0.000 to 0.001 in].

Make sure the idler shaft is properly aligned and seated against the shoulder of the internal bore.



Repeat the procedure if installing a second gear.

Remove the capscrews, replacer idler plate, and handle from the idler shaft.

Lubricate the threads and under the capscrew head with clean engine oil.

Install the washer, if applicable, and capscrew and tighten the capscrew.

Note : Not all engines will be built with a washer. The idler gear shaft without a washer will feature a 19 mm hole through the shaft.

Torque Value:

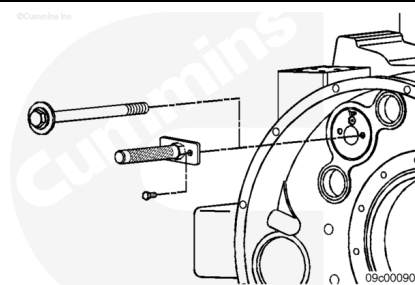
Idler Shaft Assembly With Washer

1. 100 n•m [74 ft-lb]
2. 197 n•m [145 ft-lb]

Torque Value:

Idler Shaft Assembly Without Washer

1. 135 n•m [100 ft-lb]
2. 280 n•m [207 ft-lb]



A second idler gear can be installed or an insert shaft can be used without the idler gear and roller.

Install the o-rings onto the insert shaft.

Lubricate the o-rings with clean engine oil.

Install the insert shaft into the flywheel housing hole where the cover plate will be installed (opposite pump location). Align the hole with the tapped hole in the block.

Lubricate the threads and under the capscrew head with clean engine oil.

Install the washer, if applicable, and capscrew and tighten capscrew.

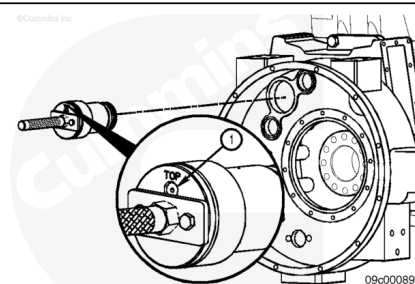
Note : Not all engines will be built with a washer. The idler gear shaft without a washer will feature a 19 mm hole through the shaft.

Torque Value:

Idler Shaft Assembly With Washer

1. 100 n•m [74 ft-lb]
2. 197 n•m [145 ft-lb]

Torque Value:



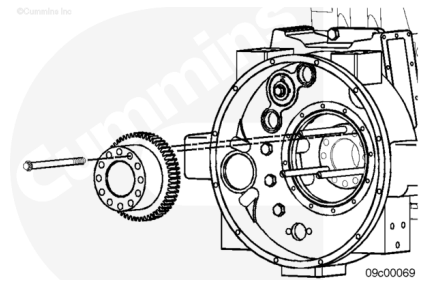
Idler Shaft Assembly Without Washer

1. 135 n•m [100 ft-lb]
2. 280 n•m [207 ft-lb]

Install the guide pins, Part Number 3164765, equally spaced through the crankshaft gear and into the crankshaft.

Rotate the idler gears, as necessary, to mesh the crankshaft gear with the idler gear(s).

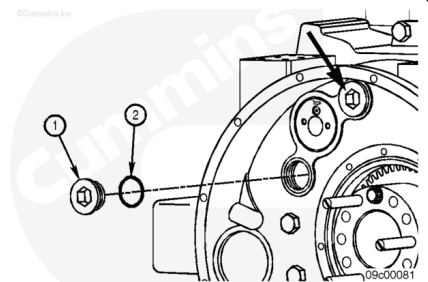
Install the crankshaft gear.



Make sure the o-rings are properly seated into the groove before installation.

Use clean engine oil to lubricate the o-rings.

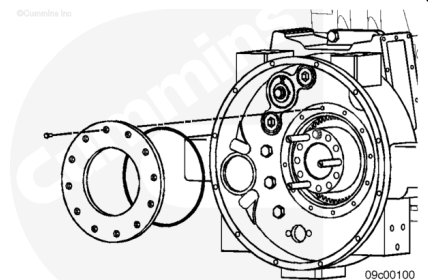
Install plugs (1) with o-rings (2) into the REPTO flywheel housing holes.



Torque Value: 28 n•m [21 ft-lb]

Install the o-ring in the REPTO flywheel housing groove.

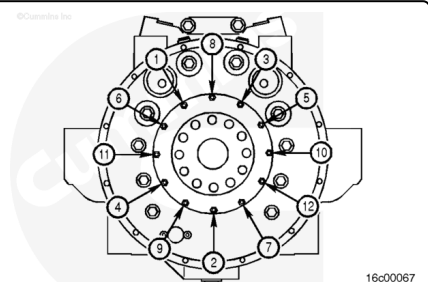
Install the seal carrier over the dowel pins on the REPTO flywheel housing.



Install the capscrews.

Tighten the capscrews in the sequence shown to secure the seal carrier.

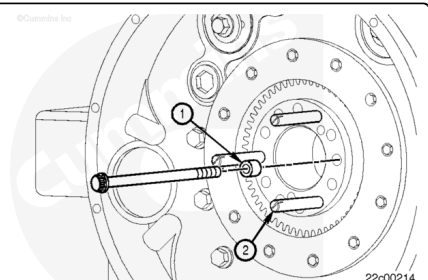
Torque Value: 28 n•m [21 ft-lb]



Install the flywheel mounting capscrews and spacers. Tighten the flywheel mounting capscrews.

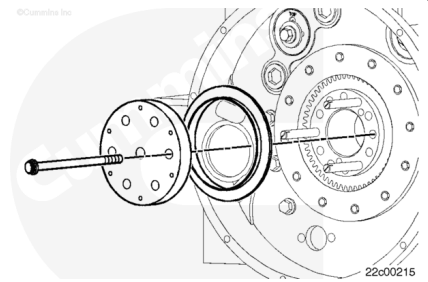
Torque Value: 60 n•m [44 ft-lb]

Remove the flywheel mounting capscrews and spacers.



⚠ CAUTION ⚠

The rear crankshaft seal must be pressed evenly into the bore. If the crankshaft and crankshaft seal is not pressed evenly into the bore, the oil seal can cause equipment damage.

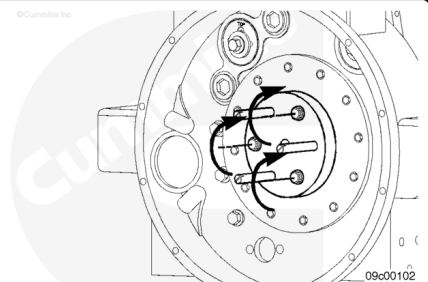


Install the rear crankshaft oil seal over the rear crankshaft seal replacer, Part Number 3164780, and onto the guide pins.

Install flywheel mounting capscrews (without the washers) through the holes in the rear crankshaft seal replacer.

⚠ CAUTION ⚠

To reduce the possibility of damage to the oil seal and crankshaft, never use a hammer or mallet to install the rear crankshaft oil seal.



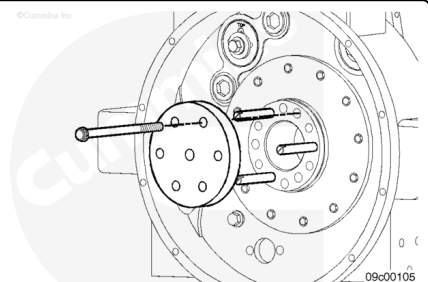
Do **not** use lubricants during the installation process. Apply a 10 percent soap solution to the outer ribbed surface of the rear crankshaft oil seal.

Tighten the flywheel mounting capscrews no more than $\frac{1}{2}$ revolution at a time, alternating capscrews in a **clockwise** pattern, until the rear crankshaft seal replacer, Part Number 3164780, seats against the REPTO flywheel housing.

Torque Value: 20 n•m [177 in-lb]

Remove the rear crankshaft seal replacer and the flywheel mounting capscrews.

Install the flywheel. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/377/377-016-005.html>)



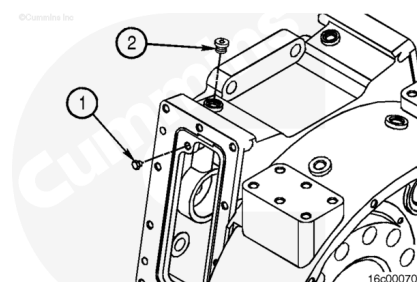
If one side of the REPTO flywheel housing cover access hole does **not** use a hydraulic housing, the opening **must** be covered in conjunction with the oil supply line hole and plugging the straight thread plug hole.

Install the 1/8-inch pipe plug (1) into the oil supply line hole. Make sure the pipe plug is flush with the mounting surface.

Torque Value: 20 n•m [177 in-lb]

Install the straight thread plug (2) into the flywheel housing hole. Make sure pipe plug is flush with the mounting surface.

Torque Value: 35 n•m [26 ft-lb]

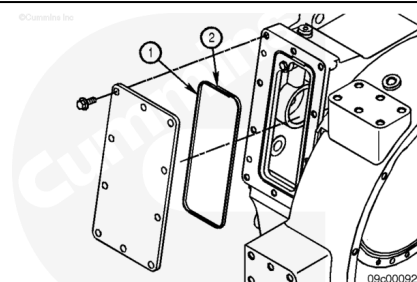


Install the diamond seal (1) on the REPTO flywheel housing cover access hole.

Apply a small amount of RTV on the two ends (2) where the diamond seal comes together and a small amount of clean engine oil on the diamond seal.

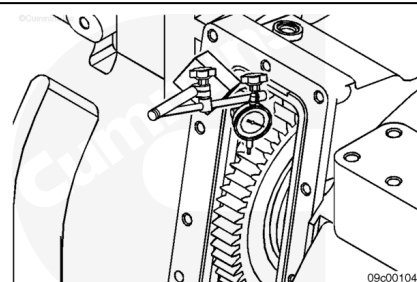
Install the cover plate to the REPTO flywheel housing cover access hole. Install the 10 capscrews and tighten.

Torque Value: 65 n•m [48 ft-lb]



Make sure the idler gear is locked in position. Use a dial gauge to check the backlash.

Idler Gear to Crankshaft Gear Backlash		
mm		in
0.0762	MIN	0.003
0.4318	MAX	0.017



If the backlash is **not** within specifications, replace the idler gear, REPTO hub gear, or both.

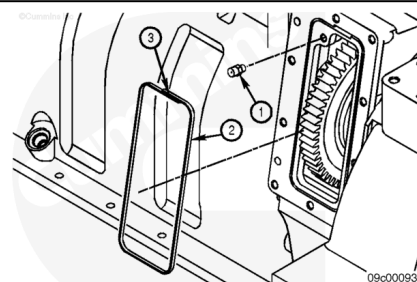
Note : The spray nozzle and diamond seal are **only** installed if a hydraulic housing is used.

Install the spray nozzle (1) into the oil supply line hole. Tighten the spray nozzle.

Torque Value: 20 n•m [15 in-lb]

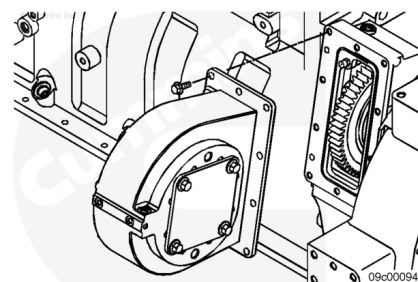
Install the diamond seal (2) on the REPTO flywheel housing cover access hole.

Apply a small amount of RTV on the two ends (3) where the diamond seal comes together and a small amount of clean engine oil on the diamond seal.



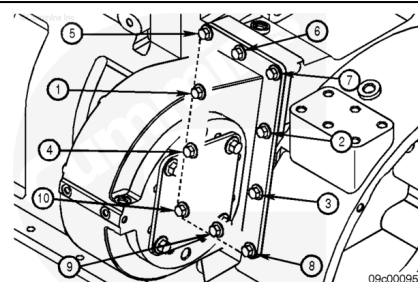
Apply clean engine oil on the gear teeth and bearings.

Install the hydraulic housing over the dowel pins. The gear inside the hydraulic housing is free to rotate and will locate the position of the idler gear as the hydraulic housing is being installed over the dowel pins.



Install the capscrews and tighten in the sequence shown.

Torque Value: 65 n•m [48 ft-lb]



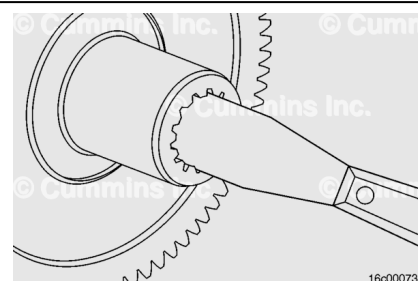
Verify that backlash exists between the Hydrive housing gear and idler gear by inserting a tool into the spline to help rotate the gear.

A small amount of backlash **must** be present, although obtaining a measurement is **not** required.

If no backlash is present, disassemble the Hydrive housing from the flywheel housing. Assemble the Hydrive housing to the opposite flywheel housing mounting pad. Check for backlash.

If no backlash is present when assembled on either flywheel housing mounting pad, the Hydrive housing **must** be replaced.

If backlash is present on **only** one side, the flywheel housing **must** be replaced.

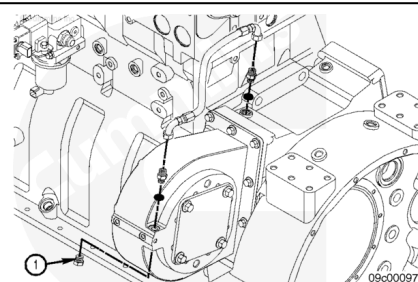


Install the threaded plug (1) in the bottom hole of the hydraulic housing. Tighten the threaded plug.

Torque Value: 32 n•m [24 ft-lb]

Install the oil line hose on the straight connectors. Tighten the hose and check for leaks.

Torque Value: 32 n•m [24 ft-lb]



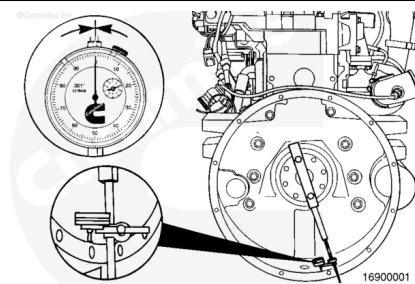
Measure

Bore Alignment

Attach a dial indicator to the crankshaft. Use mounting tool, Part Number ST-1325, and indicator, Part Number 3376050.

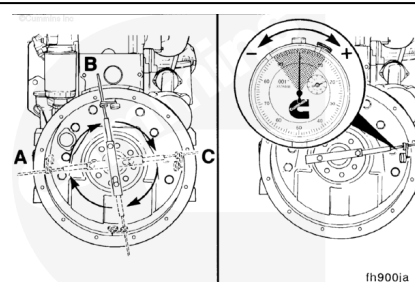
The dial indicator can be mounted by any method that holds the extension bar of the indicator rigid so it does **not** sag. If the bar sags or the indicator slips, the readings obtained will **not** be accurate.

Position the indicator in the 6 o'clock position and the gauge at zero.



Slowly rotate the crankshaft. Record the readings obtained at the 9 o'clock, 12 o'clock, and 3 o'clock positions as (a), (b), and (c) in the concentricity work sheet. Recheck zero at the 6 o'clock position.

The values for (a), (b), and (c) can be positive or negative. Reference the accompanying figure to determine the correct sign when recording these values.

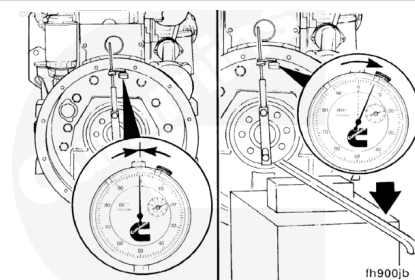


⚠ CAUTION ⚠

Do not force the crankshaft beyond the point where the bearing clearance has been removed. Do not pry against the flywheel housing. These actions can cause false bearing clearance reading.

Rotate the crankshaft until the dial indicator is at the 12-o'clock position and zero the gauge.

Use a pry bar to raise the rear of the crankshaft to its upper limit. Record the value as (d) in the concentricity work sheet. This is the vertical bearing clearance adjustment and will **always** be positive.



Use the concentricity work sheet (see the accompanying illustration) to determine the total vertical and horizontal values.

The total horizontal is equal to the 9 o'clock reading (a) minus the 3 o'clock reading (c).

Concentricity Worksheet	
9 o'clock	a = 0.004
3 o'clock	c = -0.002
Total Horizontal	a - c = .006
12 o'clock	b = .003
Bearing Clearance	d = .002
Total Vertical	b + d = .005

The total vertical is equal to the 12 o'clock reading (b) plus the bearing clearance (d).

Example:

- 6 o'clock = reference = 0
- 9 o'clock = [a] = 0.004
- 12 o'clock = [b] = 0.003
- 3 o'clock = [c] = -0.002.

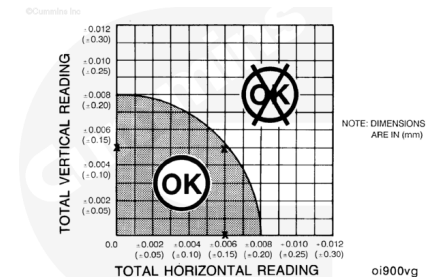
Use the work sheet and the numbers from the example. The total horizontal value equals 0.006, and the total vertical value equals 0.005.

Mark the total horizontal value on the horizontal side of the chart and the total vertical on the vertical side of the chart.

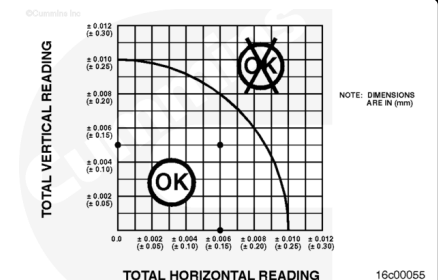
Use a straightedge to find the intersection point of the total horizontal and total vertical values. The intersection point **must** fall within the shaded area for the flywheel housing concentricity to be within specification.

Using the total horizontal and total vertical values from the previous example, the intersection point falls within the shaded area. Therefore, the flywheel housing concentricity is within specification.

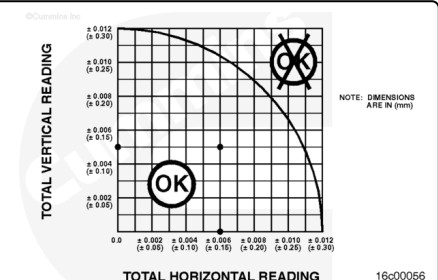
Use the accompanying chart for Society of Automotive Engineers (SAE) 1, 2, and 3 flywheel housings.



Use the accompanying chart for SAE 0 and ½ flywheel housings.



Use the accompanying chart for SAE 00 flywheel housings.



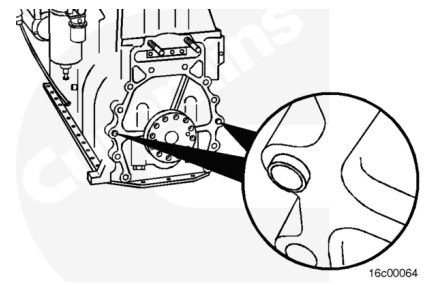
If the intersection point falls outside the shaded area, the ring dowels **must** be removed and the housing repositioned.

The ring dowels are **not** required to maintain concentricity of the housing; the clamping force of the capscrews holds the housing in place.

After the ring dowels are discarded, install the flywheel housing on the engine.

To position the housing, tighten the capscrews enough to hold the flywheel housing in place, but loose enough to enable small movement when struck lightly with a mallet.

Check the concentricity again. When concentricity is within specification, tighten the capscrews to the specified value.



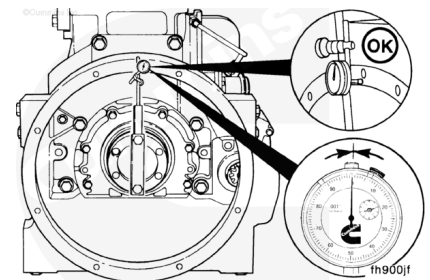
Face Alignment

⚠ CAUTION ⚠

The dial indicator tip must not enter the capscrew holes, or the gauge will be damaged.

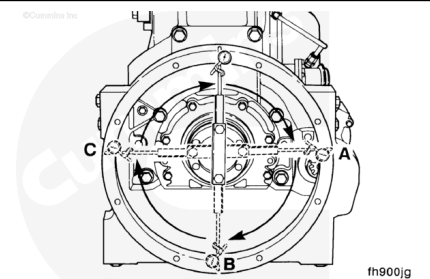
Install a dial indicator as illustrated.

The extension bar for the indicator **must** be rigid for an accurate reading. It **must not** sag. Position the indicator at the 12 o'clock position. Adjust the dial until the needle points to zero.

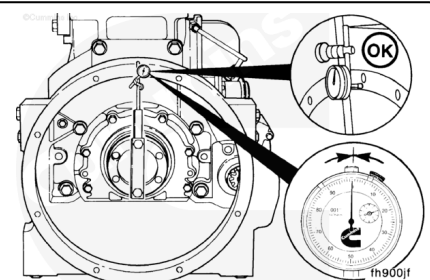


Slowly rotate the crankshaft. Record the readings at the 3 o'clock, 6 o'clock, and 9 o'clock positions.

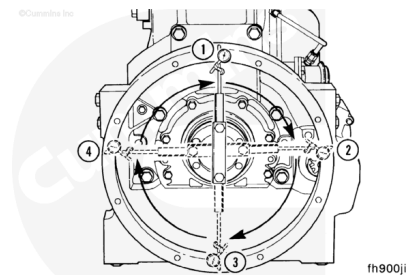
The crankshaft **must** be pushed toward the front of the engine to remove the crankshaft end clearance each time a position is measured.



Continue to rotate the crankshaft until the indicator is at the 12 o'clock position. Check the indicator to make sure that the needle points to zero. If it does **not**, the readings will be incorrect.



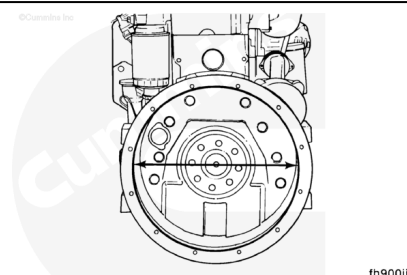
Determine the total indicator runout.



Total Indicator Runout (Example)

Position	Runout
12 o'clock	0.00 mm [0.000 in]
3 o'clock	+0.08 mm [+0.003 in]
6 o'clock	-0.05 mm [-0.002 in]
9 o'clock	+0.08 mm [+0.003 in]
Total Indicator Runout =	0.13 mm [0.005 in]

The maximum allowable total indicator reading is determined by the diameter of the housing bore. If out of specifications, replace the housing.

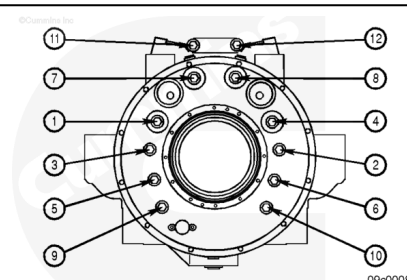


SAE Number	Bore Diameter	Total Indicator Runout Maximum
00	787.40 to 787.65 mm [30.88 to 31.00 in]	0.30 mm [0.012 in]
0	647.70 to 647.95 mm [25.50 to 25.51 in]	0.25 mm [0.010 in]
½	584.2 to 584.4 mm [23.000 to 23.008 in]	0.25 mm [0.010 in]
1	511.18 to 511.30 mm [20.125 to 20.130 in]	0.20 mm [0.008 in]
2	447.68 to 447.80 mm [17.652 to 17.630 in]	0.20 mm [0.008 in]
3	409.58 to 409.70 mm [16.125 to 16.130 in]	0.20 mm [0.008 in]

Install the remaining capscrews. Tighten all capscrews in two steps using the sequence shown.

Torque Value:

1. 100 n•m [74 ft-lb]
2. 197 n•m [145 ft-lb]



Finishing Steps

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the starting motor. Refer to Procedure 013-020 in Section 13.
(/qs3/pubsys2/xml/en/procedures/377/377-013-020-tr.html)
- Connect the batteries or air starter supply. See the equipment manufacturer service information.
- Fill the engine with clean oil. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-025.html)
- Install the transmission, clutch, and all related components. See the equipment manufacturer service information.